

Title of the Mini Project:**“ERROR DETECTION AND OPTIMIZED STREET LIGHTING MANAGEMENT”**

ABSTRACT

The proposed model revolutionizes street light systems by introducing a sophisticated infrastructure that amalgamates photodiode technology for ambient light assessment and CT coils for continuous power analysis, bolstered by GSM communication. Additionally, this system incorporates GPS technology to pinpoint detected error locations, while simultaneously pushing all data into the cloud for accessible storage and retrieval.

This smart street light system's core functionality lies in its utilization of photodiode technology to efficiently gauge ambient light conditions for automatic control of streetlight activation and deactivation. Simultaneously, CT coils continuously monitor the current flowing through the street lights, swiftly identifying fluctuations in power parameters. Upon detection, the GSM module promptly triggers notifications to operators, ensuring timely attention to malfunction detection and power irregularities.

The integration of GPS technology adds another layer of sophistication, enabling precise error localization for swift corrective action. Moreover, the system's data integration into the cloud facilitates comprehensive data storage and accessibility, enhancing its reliability and enabling optimized street lighting management.

Tools Required:

AURDINO UNO and AURDINO IDE

References:

- 1) A Comprehensive Review of Intelligent Street Lighting Systems" by H.H. Hwang et al.
- 2) <https://ieeexplore.ieee.org/document/9932734>

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